

Notes: Becker and Henderson (JPE, 2000)

Effects of Air Quality Regulations on Polluting Industries

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1 Big picture

- **Research question.** How did U.S. air-quality regulation affect the location, birth, survival, size, and investment behavior of plants in heavily polluting manufacturing industries?
- **Core setting.** The paper studies the Clean Air Act regime under which counties are classified as either in attainment or nonattainment of national air-quality standards. Nonattainment status triggers stricter equipment requirements and enforcement for polluting plants.
- **Main empirical object.** The paper follows plant births and existing plants across U.S. counties from 1963 to 1992 using Census plant-level data. The key treatment variable is county ozone nonattainment status after the 1977 Clean Air Act amendments.
- **Main finding.** Plant births in polluting industries fall sharply in nonattainment counties relative to attainment counties. The paper estimates reductions of roughly 26–45 percent, with the largest effects in industries with bigger, more visible plants.
- **Market-structure effect inside industries.** Regulation shifts activity toward smaller, less regulated single-plant firms, called nonaffiliates, and away from the more visible corporate or multiplant-firm sector in several industries.
- **Grandfathering.** Large plants that entered before regulation benefit from grandfathering provisions. Their survival probabilities rise, especially in nonattainment areas, slowing the replacement of older, dirtier capital.
- **Investment timing.** New plants in nonattainment areas appear to start larger, consistent with more up-front investment caused by costly regulatory negotiation and equipment approval. Over time, plants in attainment areas tend to catch up in size.
- **Policy trade-off.** Nonuniform regulation reduces pollution in the dirtiest areas and may move pollution toward less populated areas, but it also distorts location, plant scale, industry structure, and turnover.
- **Economic takeaway.** Environmental regulation changes more than emissions. It reshapes industrial organization by changing where firms locate, which firms enter, which plants survive, and how investment is timed.

2 Incremental contribution of the paper

- **From compliance costs to firm behavior.** Earlier work often asked whether environmental rules raise operating costs or reduce employment. Becker and Henderson study how regulation changes plant-level entry, exit, size, and investment behavior.
- **County-level regulatory variation.** The paper uses the annual county attainment/nonattainment designation as the central regulatory variable, rather than using broad state environmental indices.
- **Plant births instead of plant stocks only.** Prior evidence using plant stocks can mask opposing effects on births and deaths. This paper directly analyzes plant births, survival, and size dynamics.

- **Long plant-level panel.** The authors use the Longitudinal Research Database from the U.S. Census Bureau, linking plants and firms across Census of Manufactures years from 1963 to 1992.
- **Industry-specific analysis.** The paper focuses on major VOC and NO_x emitters that are not major polluters along other criteria, reducing confounding from other environmental programs.
- **Heterogeneity by plant and firm type.** The paper distinguishes corporate plants owned by multiplant firms from smaller nonaffiliate plants owned by single-plant firms. This is important because enforcement intensity rises with size and visibility.
- **Multiple mechanisms.** The paper identifies four channels: relocation to attainment counties, proliferation of small-scale nonaffiliate plants, grandfathering of older plants, and changes in investment timing.
- **Welfare framing.** The paper does not claim that nonuniform regulation is simply inefficient. It emphasizes a trade-off between production distortions and environmental benefits from targeting the worst polluted areas.

3 Regulatory setting and hypotheses

3.1 Clean Air Act regulation and nonattainment status

The paper focuses on ozone regulation. Ozone is produced partly by volatile organic compounds (VOC) and nitrogen oxides (NO_x). The relevant regulatory status is whether a county is in attainment or nonattainment of national ambient air-quality standards.

- **Before 1970.** Air-quality regulation was mainly state-level and weakly enforced.
- **1970 Clean Air Act amendments.** The federal government created the EPA and national ambient air-quality standards.
- **1977 amendments.** The key regime for this paper begins here. Each county is classified annually as in or out of attainment for each criterion pollutant.
- **New plants in nonattainment counties.** New polluting plants face the lowest achievable emission rate, or LAER, and may need pollution offsets.
- **New plants in attainment counties.** Large class A polluters face best available control technology, or BACT, which incorporates cost considerations and is less stringent than LAER. Small new plants in attainment counties are effectively unregulated.
- **Existing equipment.** Existing plants in nonattainment counties can face reasonably available control technology, whereas existing equipment in attainment areas is much less affected.

Interpretation. The same industry faces different effective regulation depending on where the plant locates, how large it is, when it entered, and how visible it is to regulators.

3.2 Why regulation is nonuniform

Regulation is intentionally stricter in dirtier places and for larger sources, but implementation makes the nonuniformity even stronger.

- **Spatial nonuniformity.** Nonattainment counties face stricter rules than attainment counties.
- **Size nonuniformity.** Larger plants are more likely to be class A polluters and are more likely to be inspected.
- **Age nonuniformity.** Existing plants can keep older equipment longer because of grandfathering.
- **Time nonuniformity.** Regulators tend to regulate large plants first and smaller plants later.
- **Visibility.** Multiplant corporate firms are more visible than small single-plant firms and may be easier enforcement targets.

Economic message. Even if regulators intend to reduce emissions, uneven rules create margins of avoidance: firms can change location, scale, ownership structure, and investment timing.

3.3 Main hypotheses

- **Birth relocation.** New polluting plants should become less likely to enter nonattainment counties and more likely to enter attainment counties.
- **Timing by sector.** Because regulators focus first on bigger and more visible plants, the shift away from nonattainment counties should appear earlier for corporate plants than for nonaffiliates.
- **Timing by industry.** Industries with larger plants and heavier pollution should respond earlier and more strongly.
- **Nonaffiliate expansion.** Small, less regulated firms may gain a competitive edge, shifting the industry toward nonaffiliates.
- **Grandfathering.** Older plants should survive longer, especially if they entered before the full regulatory regime and can continue operating with old equipment.
- **Investment timing.** New plants in nonattainment areas may invest more up front to avoid repeated regulatory negotiation and to keep equipment specifications consistent.

4 Data and measurement

4.1 Industry selection

The paper chooses industries that are major emitters of VOC and NO_x, but not major polluters along other air-quality dimensions. This helps isolate ozone regulation.

- **Industrial organic chemicals.** SIC 2865 and 2869 combined. This industry manufactures many VOC-related products and has large plants.
- **Metal containers.** SIC 3411 and 3412 combined. This is a major surface-coating industry.
- **Miscellaneous plastic products.** SIC 308. Plastics uses VOCs intensively in production.
- **Wood furniture.** SIC 2511. This is another surface-coating industry.

- **Excluded focus industries.** Commercial printing, gravure is dropped because it is very small before 1970. Motor vehicles are mentioned but not used in the detailed analysis because of small plant counts and disclosure limits.
- **Control industries.** The control group includes low-air-pollution industries such as apparel, mattresses and bedsprings, and leather goods.

Interpretation. The treated industries are not meant to represent all manufacturing. They are industries for which ozone-related regulation should matter strongly.

4.2 Plant and firm data

Plant-level data come from the U.S. Census Bureau’s Longitudinal Research Database.

- **Census years.** The analysis uses Census of Manufactures years: 1963, 1967, 1972, 1977, 1982, 1987, and 1992.
- **Plant identity.** Plants are followed over time using permanent plant numbers.
- **Plant birth.** A birth is a new permanent plant number, including new construction, reopening of a previously closed manufacturing plant, conversion from nonmanufacturing use, or relocation.
- **Corporate sector.** Plants owned by multiplant firms. These plants are generally much larger and more visible.
- **Nonaffiliate sector.** Plants owned by single-plant firms. These plants are smaller and often serve more local or specialized markets.
- **False births.** In 1987 and 1992, some corporate mergers or administrative restructurings can create false plant identifiers. The authors check that correcting these issues does not change the main results.

4.3 County regulation and county controls

The geographic unit is the county.

- **Regulatory status.** Since 1978, annual attainment status by county is published in the Code of Federal Regulations.
- **Early 1970s mapping.** For the early 1970s, air-quality-control-region classifications are mapped to counties, with priority 1 ozone regions treated as nonattainment.
- **County scale.** The models control for manufacturing employment outside the own industry.
- **Wages.** County manufacturing wages outside the own industry are used as a cost control.
- **Fixed effects.** County fixed effects absorb time-invariant local productivity and cost advantages that historically made some counties attractive to polluting industries.

Interpretation. Nonattainment counties are often places that historically attracted polluting firms. Fixed effects are essential; otherwise regulation could be confused with the underlying attractiveness of dirty counties.

5 Methodology and empirical specifications

5.1 Birth model

For each industry, the authors model plant births in county j and period t as a reduced-form function of county characteristics and county fixed effects:

$$B_{jt} = B(Y_{jt}, f_j + e_{jt}).$$

Because most county-period observations have zero or one births, the paper estimates a conditional Poisson model with county fixed effects. The expected number of births is

$$\lambda_{jt} = \exp(Y_{jt}a + f_j).$$

- **Outcome.** The number of plant births in a county between two Census years.
- **Key regressor.** Nonattainment status in the base period.
- **Controls.** County wages, county manufacturing scale, year dummies, and county fixed effects.
- **Interpretation.** Coefficients are proportional effects on expected births. For example, a coefficient of -0.45 is interpreted as roughly a 45 percent reduction in expected births.

5.2 Regulatory intensity

The baseline nonattainment indicator is expanded to capture intensity.

- **Monitored versus unmonitored counties.** Nonattainment counties that are actively monitored may face more serious regulatory pressure.
- **Degree of violation.** For monitored nonattainment counties, the paper uses the second-highest daily maximum ozone reading as a measure of how far the county is out of compliance.
- **Main expectation.** Birth reductions should be stronger in more intensely regulated counties.

5.3 Corporate logit model

For corporate births, the authors also estimate a conditional logit model. The idea is that a multiplant firm can compare counties nationally and choose the profit-maximizing location.

- **Unit of choice.** Each corporate birth chooses among counties.
- **Comparison to Poisson.** The logit estimates ask how nonattainment status changes the probability that a birth occurs in a county.
- **Finding.** The logit evidence broadly confirms the Poisson evidence; nonattainment reduces the probability that corporate births locate in affected counties.

5.4 Survival and grandfathering

The paper estimates plant survival models to test whether grandfathering prolongs the life of existing plants.

- **Outcome.** Survival of a plant from one Census to the next.
- **Main focus.** Corporate plants in industrial organic chemicals and plastics, where sample sizes are sufficient.
- **Grandfathering prediction.** Plants born just before regulation should have unusually high survival probabilities because they retain an equipment-cost advantage.

5.5 Plant size and investment timing

The paper studies the logarithm of real plant sales as a proxy for plant size and investment timing.

- **Specification.** Plant size is regressed on county controls, plant age categories, corporate status, nonattainment status interacted with age, year dummies, and county fixed effects.
- **Age categories.** Plants are grouped as newborn, age 5–9 years, and age 10 or more years.
- **Main coefficient.** The “all” nonattainment coefficient captures whether new plants in nonattainment areas start larger.
- **Age interactions.** Negative age interactions show whether the initial size premium fades as plants age.

Interpretation. A larger initial size in nonattainment counties is consistent with more up-front investment caused by regulatory negotiation costs and irreversible equipment choices.

6 Identification and interpretation

6.1 What the paper can identify

The paper identifies how plant-level outcomes changed in nonattainment counties relative to attainment counties after federal air-quality regulation became binding, controlling for county fixed effects and aggregate time shocks.

- The main evidence is strongest for entry location and plant births.
- The survival and size evidence supports mechanisms: grandfathering and up-front investment.
- The design is observational, but the regulatory variation is closely tied to the legal structure of the Clean Air Act.

6.2 Why fixed effects matter

Polluting plants historically chose nonattainment counties for a reason. These counties may have better market access, labor pools, suppliers, or historical industrial infrastructure.

- Without county fixed effects, nonattainment may proxy for attractive industrial locations.
- With county fixed effects, identification comes from changes within counties over time as regulation changes.
- This is why the paper treats simple cross-sectional comparisons as potentially misleading.

6.3 Control industries and regional shifts

A potential concern is that the estimated nonattainment effect captures broad regional migration away from older industrial areas rather than regulation.

- The paper controls for county manufacturing scale outside the own industry.
- It examines regional patterns and argues that shifts toward attainment areas occur within regions.
- It estimates similar models for relatively clean control industries and generally does not find comparable nonattainment effects.

Interpretation. The control-industry evidence strengthens the view that the main results reflect air-quality regulation rather than broad manufacturing trends.

6.4 Interpreting the birth coefficient

The estimated birth effects are economically large. The paper reports reductions in births in nonattainment counties of about 26–45 percent across the four main industries.

- These coefficients do not directly measure compliance costs.
- Under additional assumptions about local entry supply, the paper suggests that regulation could lower net present value for a typical new plant in a nonattainment county by roughly 13–22 percent.
- The estimates apply to heavy polluting industries and should not be generalized mechanically to all manufacturing.

6.5 Limitations

- The analysis focuses on four major polluting industries, not the entire economy.
- The paper does not compute a complete welfare effect because it does not value all health gains, relocation costs, scale distortions, and pollution spillovers.
- Some plant births may be mismeasured because of changes in permanent plant numbers, though the authors conduct checks for false births.
- Nonattainment status is an imperfect measure of the true regulatory burden because enforcement intensity varies by state, county, plant size, and time.

7 Tables and figures

7.1 Table 1 and Figure 1: Industry structure and movement out of dirty counties

Read Table 1:

- Table 1 summarizes the size and composition of four polluting industries: industrial organic chemicals, metal containers, plastics, and wood furniture.
- Corporate plants are much larger than nonaffiliate plants, especially in industrial organic chemicals and metal containers.
- Inspection rates are much higher in bigger-plant industries. In 1985–92, inspection rates are 20 percent for industrial organic chemicals and 27 percent for metal containers, versus 2.4 percent for plastics and 6.8 percent for wood furniture.
- Across the industries, the share of births in dirty counties falls from the preregulatory period to the end of the sample.
- The corporate share of births declines in the early regulatory years, consistent with stricter regulation hitting larger, more visible plants.

Read Figure 1:

- Figure 1 plots dirty counties' share of plant stocks and births over time.
- Industrial organic chemicals show a sharp drop in the birth share in dirty counties after the regulatory regime strengthens.
- Plastics and wood furniture show more gradual but sustained declines.
- The figure visually motivates the econometric birth models.

Economic message: The raw data already show a shift away from dirty counties, but the figure also shows why the paper needs econometric controls: some industries have noisy patterns and different trends.

TABLE 1
SIZE AND COMPOSITION OF INDUSTRIES

	1967-72	1977-82	1987-92
Industrial Organic Chemicals			
Stock of plants	684	865	898
Corporate	62%	58%	60%
New plants (births)	191	246	204
Corporate	42%	28%	41%
In dirty counties*	70%	57%	59%
Average real value of sales (millions of 1987 dollars):			
Corporate	87	72	95
Nonaffiliates	3.3	2.8	5.3
Average corporate employment	304	264	213
All corporate births ¹ (thousands of 1987 dollars): total real value of equipment/total employment	258	406	438
Plants inspected, 1985-92 ²	...	...	20%
1963 plants surviving to 1992	...	...	56%
Metal Containers			
Stock of plants	547	564	478
Corporate	76%	71%	65%
New plants (births)	178	135	120
Corporate	60%	39%	38%
In dirty counties*	68%	67%	63%
Average real value of sales (millions of 1987 dollars):			
Corporate	31	32	39
Nonaffiliates	3.0	2.3	3.4
Average corporate employment	179	137	116
All corporate births ¹ (thousands of 1987 dollars): total real value of equipment/total employment	96	95	130
Plants inspected, 1985-92 ²	...	...	27%
1963 plants surviving to 1992	...	...	31%
Plastics			
Stock of plants	7,608	11,630	13,073
Corporate	26%	25%	28%
New plants (births)	4,082	4,524	4,644
Corporate	21%	17%	19%
In dirty counties*	70%	67%	59%
Average real value of sales (millions of 1987 dollars):			
Corporate	10	9.7	14
Nonaffiliates	1.2	1.4	2.7
Average corporate employment	121	103	112
All corporate births ¹ (thousands of 1987 dollars): total real value of equipment/total employment	14	32	48
Plants inspected, 1985-92 ²	...	...	2.4%
1963 plants surviving to 1992	...	...	54%

Table 1, part 1: Size and composition of industries.

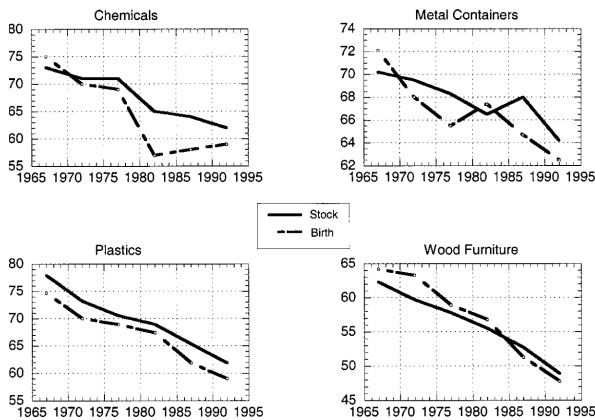


FIG. 1.—“Dirty” counties’ percentage share of births and stocks

Figure 1: Dirty counties’ percentage share of births and stocks.

TABLE 1 (Continued)

	1967-72	1977-82	1987-92
Wood Furniture			
Stock of plants	2,339	2,600	2,783
Corporate	16%	14%	11%
New plants (births)	1,300	1,318	1,279
Corporate	8.9%	4.6%	4.5%
In dirty counties*	63%	57%	48%
Average real value of sales (millions of 1987 dollars):			
Corporate	13	11	17
Nonaffiliates	1.1	.93	.95
Average corporate employment	231	229	253
All corporate births ¹ (thousands of 1987 dollars): total real value of equipment/total employment	9.5	9.6	14
Plants inspected, 1985-92 ²	...	...	6.8%
1963 plants surviving to 1992	...	...	49%

* Counties that are in nonattainment in all of 1978, 1982, and 1987.

¹ Capital stock numbers here are largely imputed by the Census Bureau.

² This ratio is the total number of plants reported in the EPA database (County Point Source Summary [AFPS49]) as having at least one inspection from 1985 to 1992 divided by the 1987 stock of plants. For the four industries, the ratios of plants that were class A polluters in that sample for VOC or NO_x emissions relative to the 1987 stock are, respectively, 12 percent, 14 percent, 0.7 percent, and 5.0 percent, and the ratios of class A polluters relative to the 1987 stock of corporate plants are 20 percent, 21 percent, 2.6 percent, and 42 percent.

Table 1, part 2: Continued industry size and composition.

TABLE 2
BIRTHS: OVERALL NONATTAINMENT EFFECTS

	Industrial Organic Chemicals	Metal Containers	Plastics	Wood Furniture
A. Basic Formulation				
Nonattainment status (1978, 1982, 1987)	-.45**	-.27*	-.26**	-.29**
B. Formulation Allowing for Intensity of Regulation				
Nonattainment:				
County not monitored	-.51**	-.46*	-.19**	-.23**
County monitored	-.93**	-.31	-.61**	-.56**
In (second-highest daily maximum)—if monitored and nonattainment	-.26	-.03	-.17	-.13
In (second-highest daily maximum for 1987-92), corporate plants	-1.74**	-.48	-.44**	-1.83**

NOTE.—Covariates in col. 1 of tables A1–A4 are wages, county scale, nonattainment status, and year dummies. Fixed effects are conditioned out.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

Table 2: Overall nonattainment effects on births.

7.2 Table 2 and Table 3: Birth effects and timing by sector

Read Table 2:

- Table 2 reports the baseline birth effects of nonattainment status.
- Nonattainment reduces expected births by 45 percent in industrial organic chemicals, 27 percent in metal containers, 26 percent in plastics, and 29 percent in wood furniture.
- The effects are stronger in many cases for monitored nonattainment counties, which suggests that regulatory intensity matters.
- For corporate plants in 1987–92, the degree of nonattainment has especially large negative effects in several industries.

Read Table 3:

- Table 3 splits birth effects by year and by sector, separating corporate and nonaffiliate plants.
- Within industries, nonattainment effects generally appear first for corporate plants and later for nonaffiliates.
- Across industries, the shift begins earlier in bigger-plant, heavier-polluting industries such as industrial organic chemicals and metal containers.
- The time dummies indicate that nonaffiliate births expand relative to corporate births in several industries after regulation.
- The conditional logit estimates for corporate births in panel B broadly reinforce the Poisson results.

Economic message: Regulation does not merely reduce entry uniformly. It changes the timing and composition of entry, first discouraging large corporate plants and then pushing smaller plants toward attainment areas.

TABLE 3
BIRTHS: NONATTAINMENT STATUS EFFECTS BY YEAR AND SECTOR

YEAR	INDUSTRIAL ORGANIC CHEMICALS		METAL CONTAINERS		PLASTICS		WOOD FURNITURE	
	Corporate	Nonaffiliated	Corporate	Nonaffiliated	Corporate	Nonaffiliated	Corporate	Nonaffiliated
A. Results from Cols. 3 and 4 of Tables A1–A4								
1972 (for 1972–77 births)	-.49*	.57**	-.16	-.09	-.03	.14**	.03	.02
1978 (for 1977–82 births)	-.29	-.53*	-.66*	.05	-.37**	.05	.05	-.11
1982 (for 1982–87 births)	-.65**	-.60**	.14	-.59*	-.41**	-.18**	-.55**	-.27**
1987 (for 1987–92 births)	-.34	-.25	-.14	-.68**	-.45**	-.35**	-.32*	-.45**
Time dummies:								
1972–77	.10	-.02	-.32	.05	.13	-.01	-.18	.33**
1977–82	.10	.78**	-.15	.13	.12	.04	-.54*	.11
1982–87	.60**	.39*	-.80**	.65**	.39**	.07	-.04	.32**
1987–92	.10	.47*	-.88**	.51**	.24*	.22**	-.28	.21**
B. Results from Logit Formulation in the Corporate Sector								
1972	-.46*		-.24		.15***		.02	
1978	-.23		-.64*		-.34***		.07	
1982	-.58**		-.01		-.48***		-.56**	
1987	-.35**		-.10		-.49***		-.50**	

NOTE.—Coefficients are wages, county scale, nonattainment status by year, and year dummies. Fixed effects are conditioned out.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Because of computational constraints (the number of county dummy variables), these coefficients are obtained from 11 estimations of 10 percent random samples from plastics data. For 1978, 1982, and 1987, these coefficients are significantly negative at the 10 percent level in four, six, and eight cases, respectively. For 1982 and 1987, the coefficients range from zero to $-.86$.

Table 3: Nonattainment effects by year and sector.

TABLE 4
SURVIVAL PROBABILITY: CORPORATE PLANTS IN A
TYPICAL COUNTY

	Industrial Organic Chemicals	Plastics
Births 1963–67	.57	.51
Births 1967–72:		
Attainment areas	.98**	.57
Nonattainment areas	.94	.57
Births 1972–77:		
Attainment areas	.10	.31**
Nonattainment areas	.71*	.49**
Births 1977–82:		
Attainment areas	.39	.27**
Nonattainment areas	.56	.50**
Births 1982–87:		
Attainment areas	.82	.50
Nonattainment areas	.88	.50

NOTE.—Estimated probits for industrial organic chemicals ($N = 613$) are (standard errors are in parentheses):

$$\begin{aligned} \text{prob}(\text{survive}) = & 2.13 + .394 \ln(\text{all other manufacturing employment}) \\ & (3.94) \quad (.321) \\ & - 3.06 \ln(\text{real wage}) + .455 \text{ corporate dummy} \\ & (1.24) \quad (.128) \\ & + 2.05 \text{ born } 1967\text{--}72 - 1.43 \text{ born } 1972\text{--}77 \\ & (.715) \quad (1.01) \\ & - .476 \text{ born } 1977\text{--}82 + .737 \text{ born } 1982\text{--}87 \\ & (.635) \quad (.493) \\ & - .713 \text{ nonattainment } 1972 \text{ (for } 1967\text{--}72 \text{ births)} \\ & (.547) \\ & + 1.81 \text{ nonattainment } 1977 + .466 \text{ nonattainment } 1982 \\ & (1.02) \quad (.645) \\ & + .278 \text{ nonattainment } 1987 + \text{county dummy variables} \\ & (.434) \end{aligned}$$

For plastics ($N = 3,158$):

$$\begin{aligned} \text{prob}(\text{survive}) = & .273 - .042 \ln(\text{all other manufacturing employment}) \\ & (1.84) \quad (.147) \\ & + .209 \ln(\text{real wage}) + .158 \text{ born } 1967\text{--}72 \\ & (.506) \quad (.192) \\ & - .510 \text{ born } 1972\text{--}77 - .639 \text{ born } 1977\text{--}82 \\ & (.217) \quad (.238) \\ & - .031 \text{ born } 1982\text{--}87 + .0018 \text{ nonattainment } 1972 \\ & (.216) \quad (.143) \\ & + .465 \text{ nonattainment } 1977 + .678 \text{ nonattainment } 1982 \\ & (.199) \quad (.197) \\ & + .013 \text{ nonattainment } 1987 + \text{county dummy variables} \\ & (.156) \end{aligned}$$

* Calculated probability based on a dummy variable significant at the 10 percent level.

** Calculated probability based on a dummy variable significant at the 5 percent level.

Table 4: Survival probability for corporate plants in a typical county.

7.3 Table 4 and Tables 5–6: Grandfathering, plant size, and investment timing

Read Table 4:

- Table 4 reports survival probabilities for corporate plants in industrial organic chemicals and plastics.
- For industrial organic chemicals, plants born in 1967–72 have very high survival probabilities in both attainment and nonattainment areas.

- This pattern is consistent with grandfathering: plants that entered before stronger regulation could continue operating with older equipment.
- In the early regulatory years, survival differs between attainment and nonattainment areas, but by 1982–87 births the probabilities converge again.

Read Table 5:

- Table 5 studies determinants of total real plant sales, a measure of plant size.
- Age effects are large: plants aged 5–9 and 10 or more years are substantially larger than newborn plants.
- Corporate plants are much larger than noncorporate plants.
- In all industries except metal containers, new plants in nonattainment counties are significantly larger than comparable new plants in attainment counties.
- Negative age interactions imply that the nonattainment size premium diminishes as plants age.

Read Table 6:

- Table 6 allows nonattainment-size effects to vary over time.
- In 1986 and 1991, new plants in nonattainment counties are especially large in industrial organic chemicals, plastics, and wood furniture.
- The negative age interactions show that plants in attainment counties catch up over time.
- The pattern supports the view that plants in nonattainment areas make larger up-front investments rather than phasing investment gradually.

Economic message: Regulation changes the life cycle of plants. It keeps older regulated plants alive and makes new plants in regulated areas start at a larger, more cautious scale.

TABLE 5
SIZE DETERMINANTS: TOTAL REAL VALUE OF SALES

	Industrial Organic Chemicals	Metal Containers	Plastics	Wood Furniture
ln(all other manufacturing)	-.075 (.129)	.256* (.149)	-.020 (.036)	-.085 (.088)
ln(real wage)	-.071 (.326)	-.576 (.402)	.040 (.108)	-.451* (.245)
Age 5–9 years	.755** (.125)	.643** (.112)	.570** (.023)	.554** (.056)
Age 10+ years	1.41** (.115)	1.03** (.110)	.950** (.022)	1.07** (.052)
Dummy corporation	1.34** (.079)	1.27** (.112)	1.16** (.015)	1.30** (.050)
Nonattainment, 1981, 1986, 1991:				
All	.685** (.175)	-.013 (.181)	.250** (.037)	.482** (.080)
Age 5–9 years	-.736* (.196)	-.158 (.211)	-.191** (.037)	-.268** (.088)
Age 10+ years	-.572** (.159)	-.202 (.173)	-.272** (.032)	-.565** (.080)
Year dummies:				
1977	-.148 (.148)	-.097 (.090)	.035 (.027)	-.023 (.049)
1982	-.477** (.161)	-.219** (.102)	-.079** (.032)	-.330** (.063)
1987	-.097 (.115)	-.046 (.098)	.367** (.032)	.073 (.065)
1992	-.139 (.122)	.077 (.108)	.308** (.033)	-.109* (.066)
Adjusted R^2	.535	.488	.372	.530
Observations	2,858	2,024	32,024	6,046
Counties	319	261	1,321	635

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table 5: Size determinants using total real value of sales.

TABLE 6
TOTAL REAL SALES: TIME-VARYING NONATTAINMENT STATUS

	Industrial Organic Chemicals	Metal Containers	Plastics	Wood Furniture
1981 nonattainment:				
All	.351 (.231)	-.453* (.257)	.134** (.052)	.429** (.102)
Age 5–9	-.417 (.275)	.339 (.299)	-.124** (.054)	-.290** (.133)
Age 10+	-.084 (.211)	.306 (.252)	-.070 (.047)	-.595** (.112)
1986 nonattainment:				
All	.859** (.231)	.241 (.252)	.423** (.049)	.552** (.106)
Age 5–9	-.910** (.279)	-.684* (.365)	-.305** (.053)	-.268** (.125)
Age 10+	-.784** (.222)	-.501** (.246)	-.482** (.046)	-.577** (.118)
1991 nonattainment:				
All	.962** (.239)	.330 (.287)	.219** (.050)	.484** (.114)
Age 5–9	-.964** (.321)	-.387 (.347)	-.156** (.056)	-.246** (.141)
Age 10+	-.980** (.227)	-.619** (.285)	-.283** (.047)	-.528** (.122)

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table 6: Time-varying nonattainment effects on total real sales.

7.4 Appendix Tables A1–A4: Conditional Poisson birth models

Read Appendix Tables A1–A4:

- The appendix reports the detailed conditional Poisson birth regressions for the four main industries.
- Table A1 covers industrial organic chemicals, Table A2 covers metal containers, Table A3 covers plastics, and Table A4 covers wood furniture.
- The first column in each table gives the baseline all-birth effect of nonattainment status.
- Later columns split births by corporate and nonaffiliate sectors and allow nonattainment effects to vary by year.
- The appendix makes transparent which industry-year-sector cells drive the summary results in Tables 2 and 3.

Economic message: The core entry result is not a single pooled coefficient. It appears across multiple industries, with predictable differences by plant size, sector, and phase of regulatory implementation.

TABLE A1
BIRTHS: INDUSTRIAL ORGANIC CHEMICALS

	ALL BIRTHS (N = 415)		CORPORATE BIRTHS (N = 212)	NONAFFILIATE BIRTHS (N = 316)
	(1)	(2)	(3)	(4)
ln(manufacturing employment)	.463** (.142)	.466** (.141)	.424** (.214)	.468** (.195)
ln(real wage)	-.834* (.466)	-.854* (.469)	.306 (.736)	-1.52** (.644)
Nonattainment: 1978, 1982, 1987	-.450** (.132)			
1972 (for 1972–77 births)		.097 (.192)	-.487* (.271)	.566** (.268)
1978 (for 1977–82 births)		-.456** (.192)	-.292 (.333)	-.532** (.251)
1982 (for 1982–87 births)		-.616** (.187)	-.647** (.286)	-.603** (.247)
1987 (for 1987–92 births)		-.279 (.192)	-.340 (.296)	-.250 (.251)
Period:				
1963–67	-.223 (.138)	-.226 (.138)	.110 (.207)	-.454** (.176)
1972–77	.166 (.217)	.105 (.258)	.104 (.351)	-.021 (.375)
1977–82	.551** (.142)	.555** (.182)	.095 (.301)	.783** (.241)
1982–87	.369** (.141)	.483** (.167)	.599** (.257)	.389* (.222)
1987–92	.437** (.185)	.334* (.199)	.095 (.318)	.469* (.268)
T	6	6	6	6
Pseudo R ²	.615	.615	.450	.453
Robust Hausman statistic	16.77	21.01	11.71	16.46

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table A1: Births in industrial organic chemicals.

TABLE A2
BIRTHS: METAL CONTAINERS

	ALL BIRTHS (N = 334)		CORPORATE BIRTHS (N = 215)	NONAFFILIATE BIRTHS (N = 212)
	(1)	(2)	(3)	(4)
ln(manufacturing employment)	.327** (.161)	.325** (.161)	.576** (.237)	.145 (.223)
ln(real wage)	.278 (.663)	.266 (.660)	1.05 (1.08)	.227 (.873)
Nonattainment: 1978, 1982, 1987	-.272* (.165)			
1972 (for 1972–77 births)		-.116 (.187)	-.159 (.261)	-.092 (.281)
1978 (for 1977–82 births)		-.232 (.268)	-.658* (.371)	.051 (.422)
1982 (for 1982–87 births)		-.240 (.232)	.142 (.377)	-.590* (.316)
1987 (for 1987–92 births)		-.371* (.221)	-.135 (.344)	-.683** (.291)
Period:				
1963–67	-.387** (.147)	-.389** (.147)	-.115 (.183)	-.730** (.236)
1972–77	-.216* (.122)	-.136 (.167)	-.316 (.231)	.054 (.259)
1977–82	-.035 (.196)	-.067 (.260)	-.148 (.342)	.132 (.407)
1982–87	-.051 (.170)	-.074 (.220)	-.799** (.344)	.647** (.289)
1987–92	-.247 (.167)	-.183 (.179)	-.881** (.271)	.512** (.259)
T	6	6	6	6
Pseudo R ²	.534	.535	.352	.446
Robust Hausman statistic	8.05	8.53	6.41	9.41

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table A2: Births in metal containers.

TABLE A3
BIRTHS: MISCELLANEOUS PLASTICS PRODUCTS

	ALL BIRTHS (N = 1,575)		CORPORATE BIRTHS (N = 1,001)	NONAFFILIATE BIRTHS (N = 1,443)
	(1)	(2)	(3)	(4)
ln(manufacturing employment)	.392** (.059)	.399** (.058)	.337** (.076)	.416** (.063)
ln(real wage)	.192 (.193)	.181 (.195)	.069 (.277)	.245 (.206)
Nonattainment: 1978, 1982, 1987	-.261** (.047)			
1972 (for 1972–77 births)		.104** (.039)	-.034 (.083)	.142** (.044)
1978 (for 1977–82 births)		-.023 (.049)	-.366** (.100)	.050 (.057)
1982 (for 1982–87 births)		-.245** (.057)	-.414** (.097)	-.184** (.062)
1987 (for 1987–92 births)		-.367** (.065)	-.453** (.098)	-.351** (.068)
Period:				
1963–67	-.591** (.066)	-.593** (.066)	-.457** (.082)	-.626** (.073)
1972–77	.099 (.063)	.033 (.072)	.131 (.116)	-.006 (.076)
1977–82	.253** (.059)	.062 (.061)	.123 (.109)	.040 (.068)
1982–87	.170** (.067)	.158** (.075)	.385** (.121)	.074 (.080)
1987–92	.165* (.096)	.232** (.093)	.244* (.146)	.218** (.097)
T	6	6	6	6
Pseudo R ²	.931	.937	.810	.936
Robust Hausman statistic	15.35	16.54	12.71	17.46

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table A3: Births in miscellaneous plastics products.

TABLE A4
BIRTHS: WOOD FURNITURE

	ALL BIRTHS (N = 1,226)		CORPORATE BIRTHS (N = 255)	NONAFFILIATE BIRTHS (N = 1,185)
	(1)	(2)	(3)	(4)
ln(manufacturing employment)	.173** (.060)	.177** (.060)	-.071 (.216)	.184** (.061)
ln(real wage)	-.018 (.225)	-.012 (.225)	-.649 (.822)	.033 (.228)
Nonattainment: 1978, 1982, 1987	-.289** (.065)			
1972 (for 1972–77 births)		.028 (.067)	.027 (.278)	.023 (.071)
1978 (for 1977–82 births)		-.095 (.084)	.050 (.311)	-.111 (.089)
1982 (for 1982–87 births)		-.276** (.089)	-.547** (.299)	-.267** (.090)
1987 (for 1987–92 births)		-.442** (.082)	-.520* (.307)	-.445** (.085)
Period:				
1963–67	-.142** (.045)	-.141** (.045)	-.846** (.194)	-.094** (.046)
1972–77	.307** (.048)	.290** (.068)	-.184 (.224)	.328** (.071)
1977–82	.195** (.075)	.056 (.085)	-.542* (.299)	.105 (.090)
1982–87	.304** (.081)	.293** (.080)	-.040 (.267)	.323** (.081)
1987–92	.094 (.079)	.172* (.088)	-.280 (.297)	.210** (.089)
T	6	6	6	6
Pseudo R ²	.926	.926	.423	.924
Robust Hausman statistic	17.93	17.03	10.41	16.59

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table A4: Births in wood furniture.

8 Short summary

- The paper studies unintended effects of U.S. air-quality regulation on major polluting manufacturing industries.
- The key regulatory variable is county ozone nonattainment status after the 1977 Clean Air Act amendments.
- Nonattainment areas face stricter equipment requirements and more intensive enforcement, especially for large plants.
- Using plant-level Census data from 1963 to 1992, the authors show that births of polluting plants fall sharply in nonattainment counties.
- The estimated birth reductions range from about 26 percent to 45 percent, depending on the industry.
- Larger and more visible corporate plants respond earlier than smaller nonaffiliate plants.
- Regulation shifts some activity toward smaller, less regulated single-plant firms.
- Grandfathering increases survival rates of older plants, especially those born before the stronger regulatory regime.
- New plants in nonattainment counties tend to start larger, consistent with more up-front investment and less phased-in growth.
- The policy reduces pollution pressure in dirty areas but creates production and organizational distortions.

9 Conclusion

9.1 Core conclusion

Becker and Henderson show that air-quality regulation substantially changed the geography and structure of polluting industries. Nonattainment status discouraged births in regulated counties, shifted entry toward attainment counties, gave an advantage to smaller and less visible firms, prolonged the life of grandfathered older plants, and altered investment timing.

9.2 Economic intuition

The Clean Air Act made polluting activity more expensive in some places than in others. Firms responded like optimizing economic agents. New plants avoided high-cost regulatory environments; small plants benefited from weaker enforcement; old plants survived longer because they had already installed capital; and new plants in regulated areas invested more up front to avoid repeated costly negotiations.

9.3 Incremental contribution revisited

The paper's main contribution is to open the black box of environmental regulation's real effects. Instead of only asking whether regulation improves air quality or raises costs, it shows how nonuniform regulation changes entry, exit, scale, location, and market structure at the plant level.

9.4 Implications

The results imply that environmental policy should be evaluated not only by emissions reductions but also by induced changes in industrial organization. A policy that targets the dirtiest counties and largest plants may be administratively sensible and environmentally valuable, but it also creates incentives to relocate production, operate at smaller scale, and keep old capital alive.

9.5 Limitations

The paper does not provide a full welfare calculation. It cannot fully quantify health benefits, productivity losses, relocation costs, or the environmental costs of shifting pollution to cleaner and often less populated areas. The analysis is also concentrated on major polluting industries, so the quantitative estimates are not meant to apply to all manufacturing.

9.6 Takeaway

The enduring takeaway is that regulation affects margins beyond the regulated outcome. Air-quality rules did not merely reduce emissions; they changed where plants were born, which firms expanded, which plants survived, and how investment was staged. The design of regulation therefore shapes both environmental outcomes and industrial structure.